

Practice Intermediate Algebra Test IV (Version 2)

Name
Date
Course

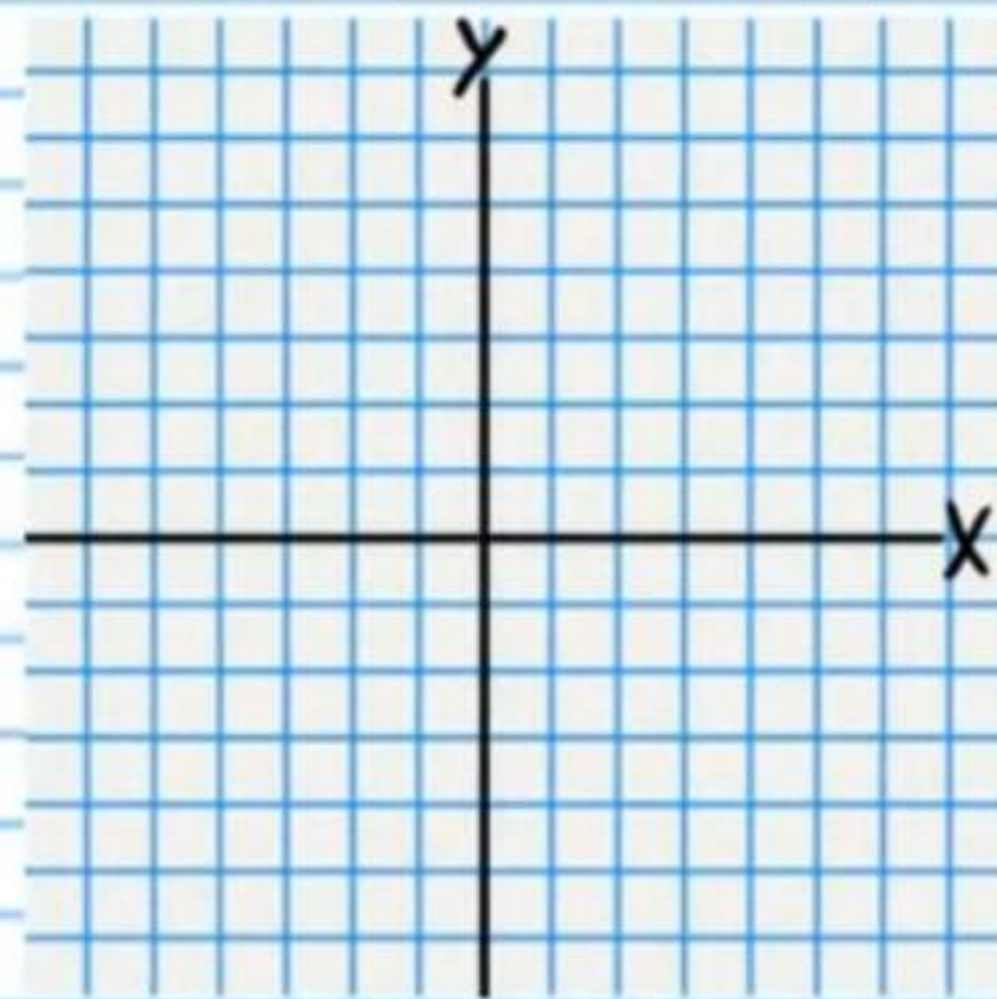
- I) Write your name, the date, and the course title (i.e. Intermediate Algebra).
- II) Show all work.
- III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) You will need a calculator for this test.
- V) Each problem is worth four points for a total of 100 points.

Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation.

① $-y = x^2$

Parent Relation:

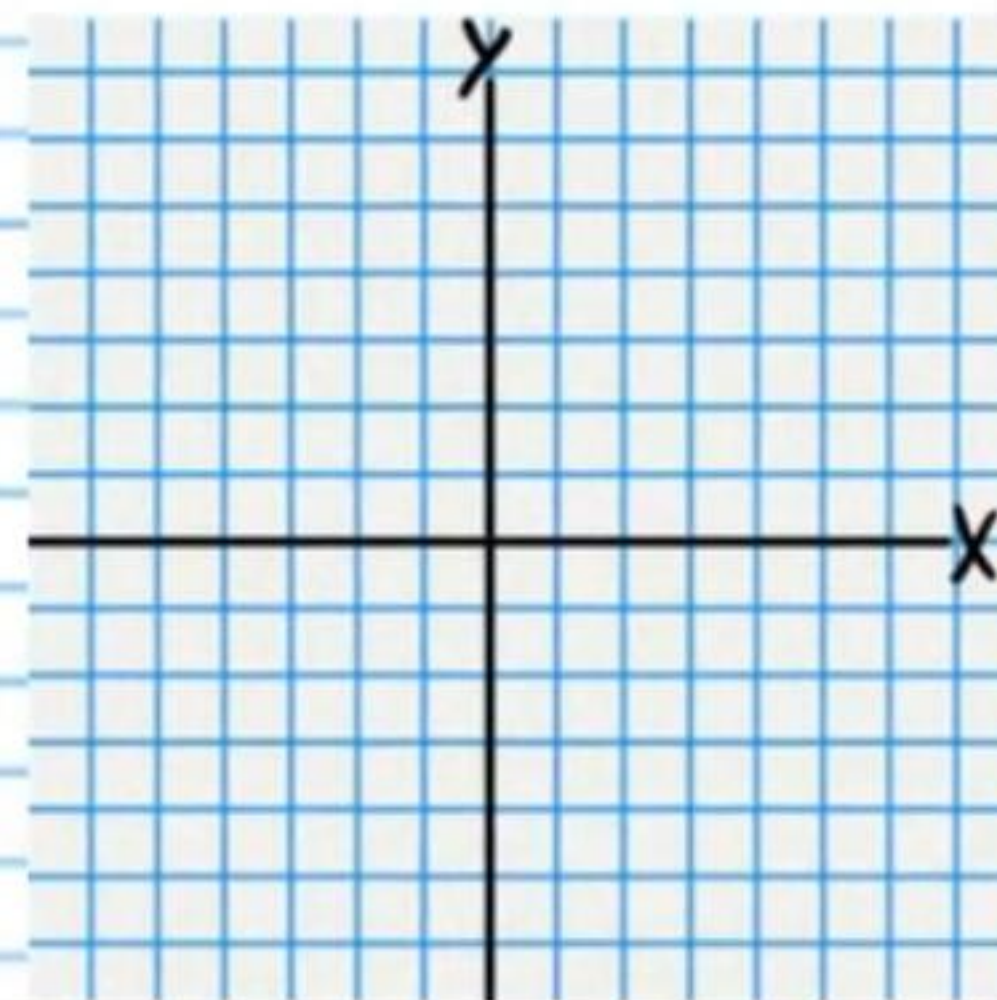
Parent Equation:



② $(-x)^3 = y$

Parent Relation:

Parent Equation:



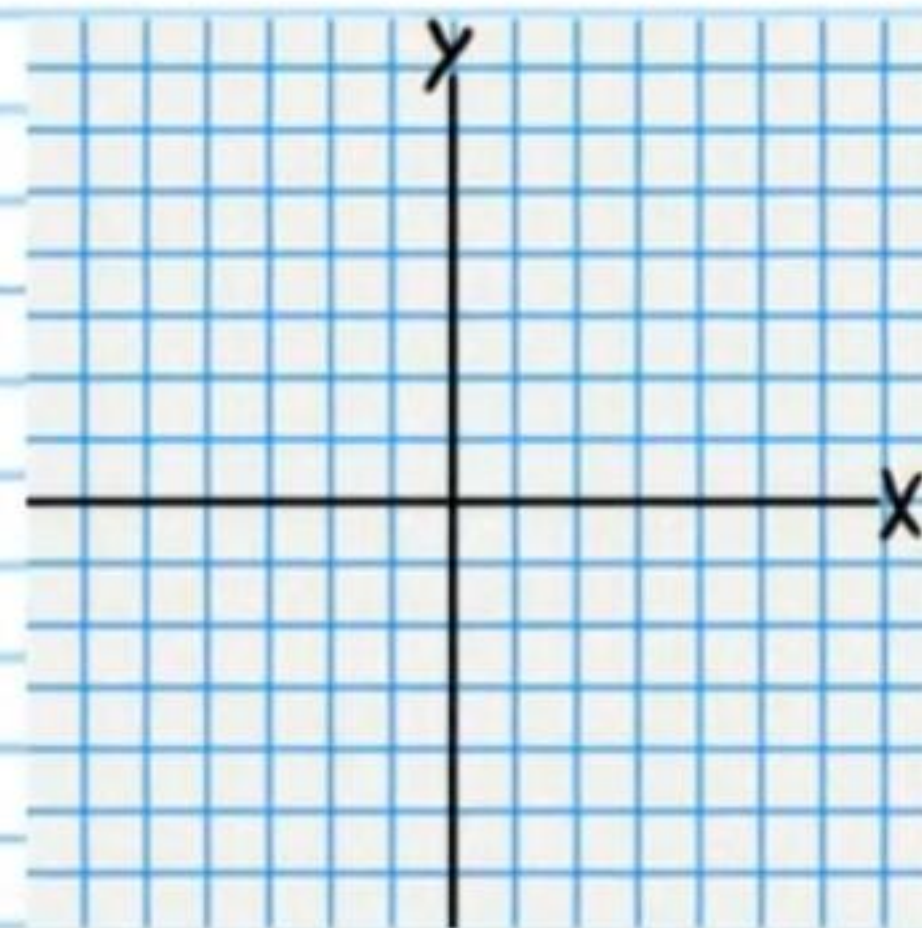
Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation. Draw the asymptote(s) on the graph as dotted lines and write their equation(s). If applicable, identify the type of dilation involved.

③ $y - 4 = \log_2(x - 2)$

Parent Relation:

Parent Equation:

Asymptote Equation:



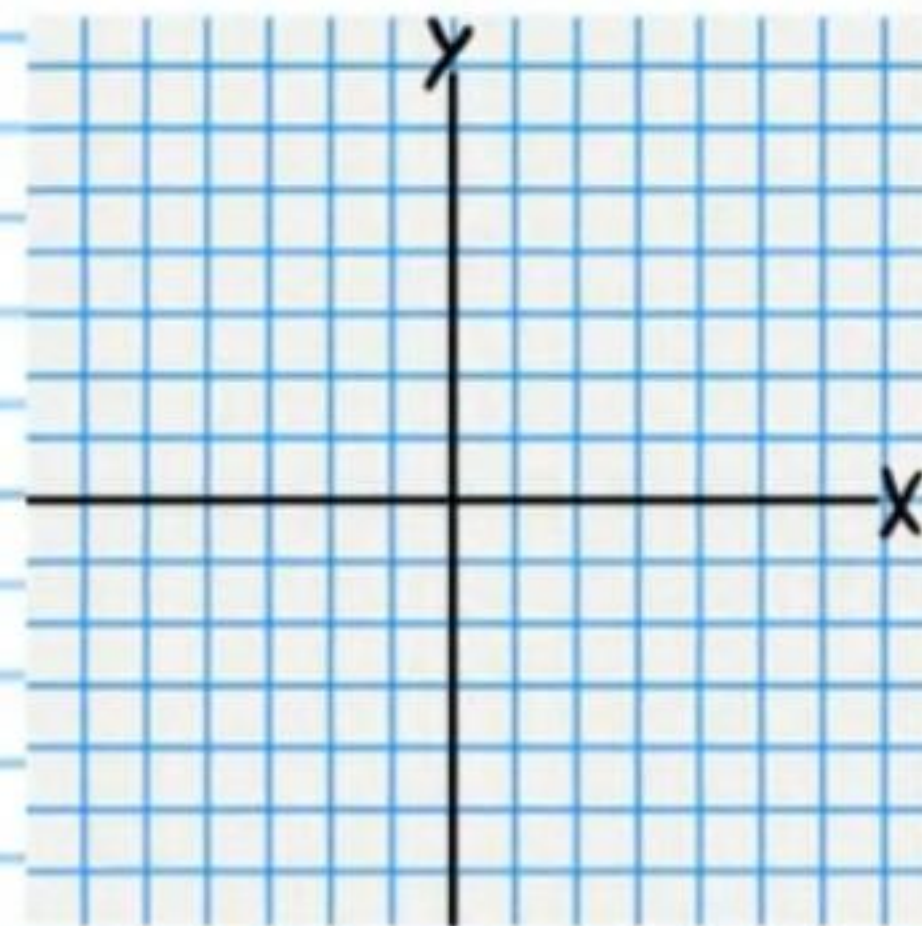
④ $y = \frac{1}{2x}$

Parent Relation:

Parent Equation:

Dilation:

Asymptote Equations:



Change the form of each equation below so that it can be graphed with the transformation method. Draw the parent graph. Then draw each transformation, and write the equation for each graph. Draw any asymptotes as dotted lines.

⑤ $4|x| = y$

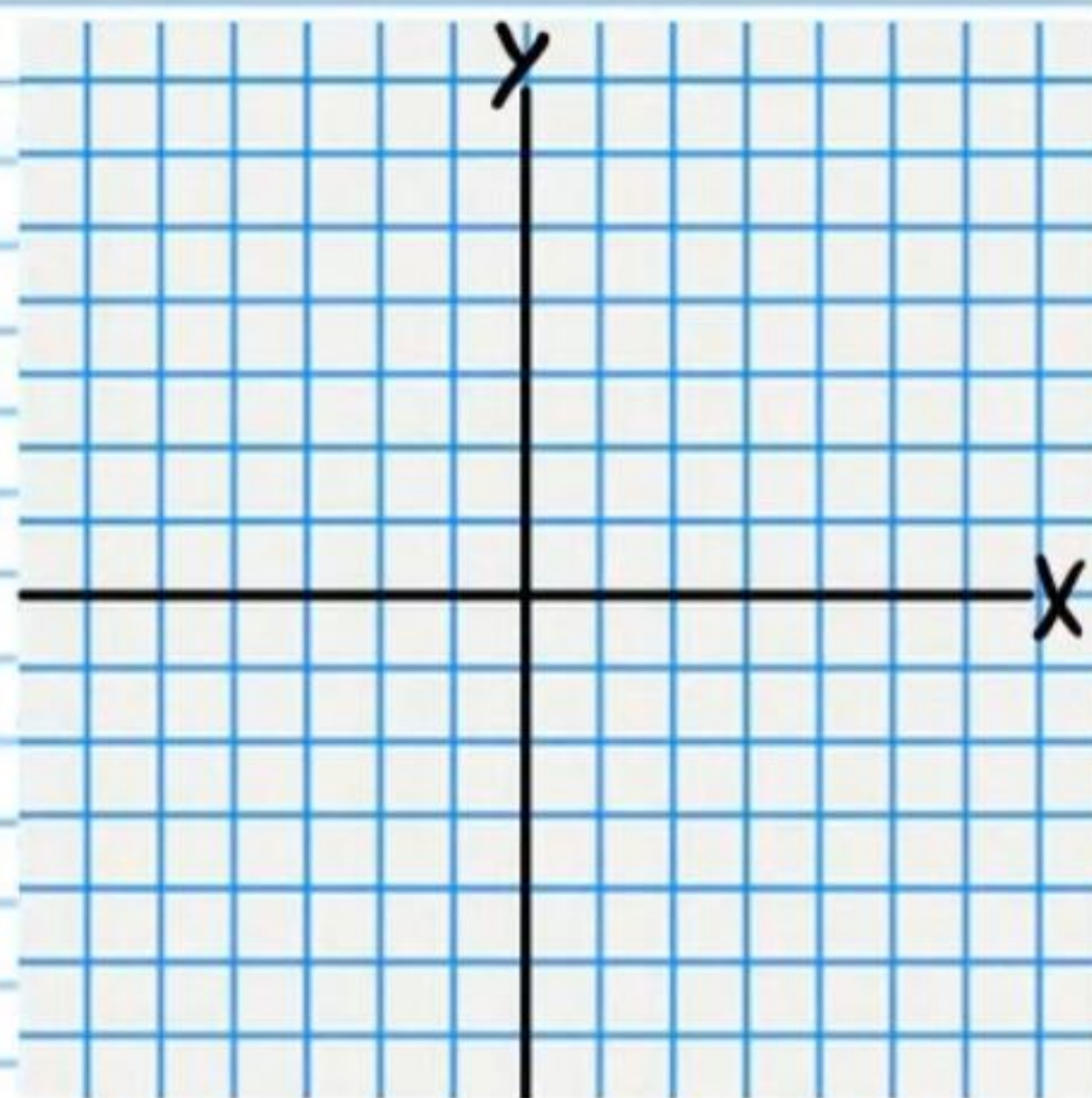
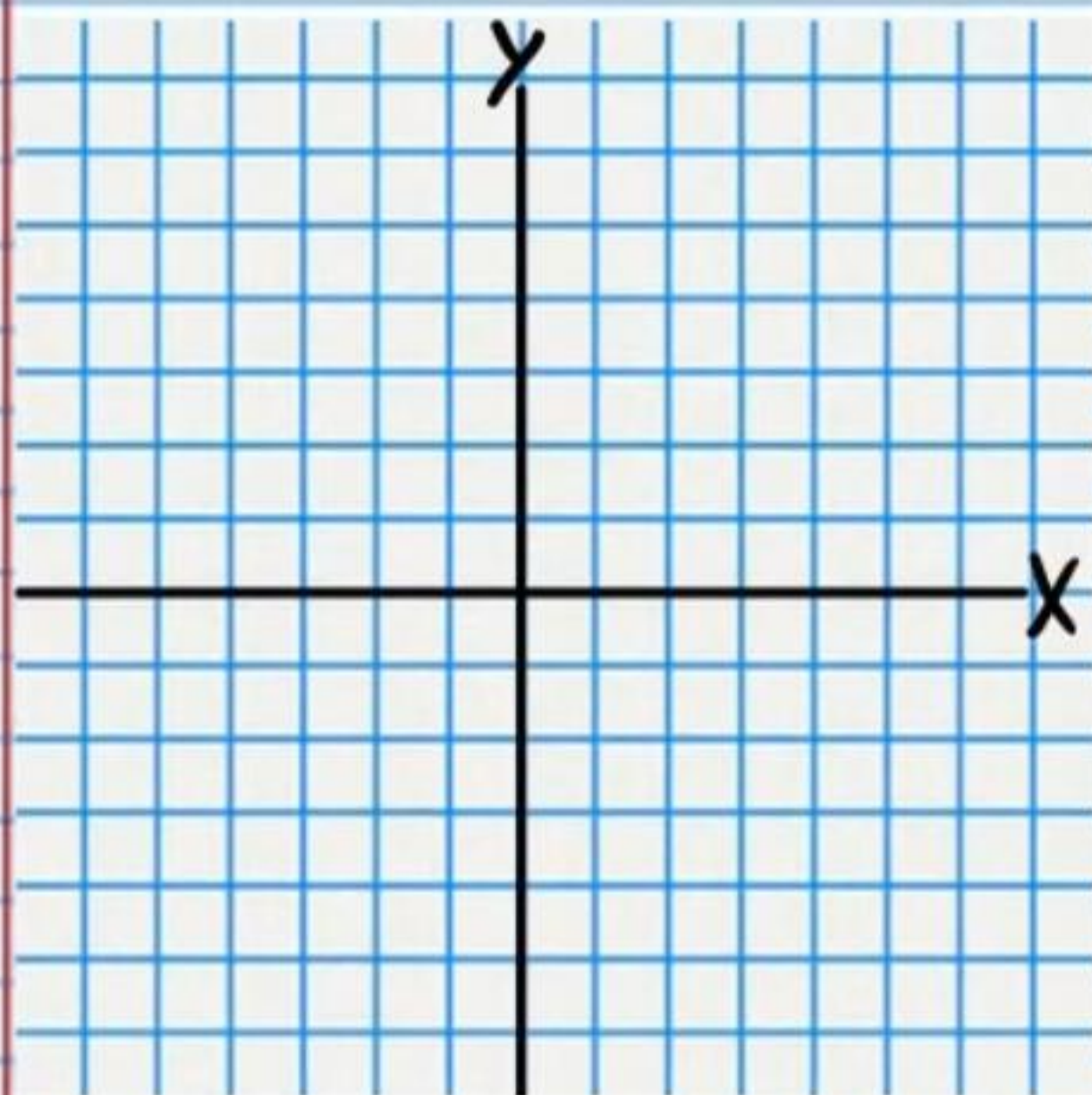
Change the form of the equation.



Altered Form:

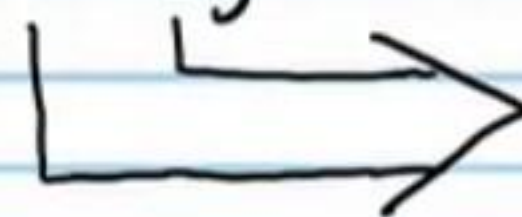
Parent Relation

Transformation



⑥ $y = 3(2^{x+3}) - 3$

Change the form of the equation.

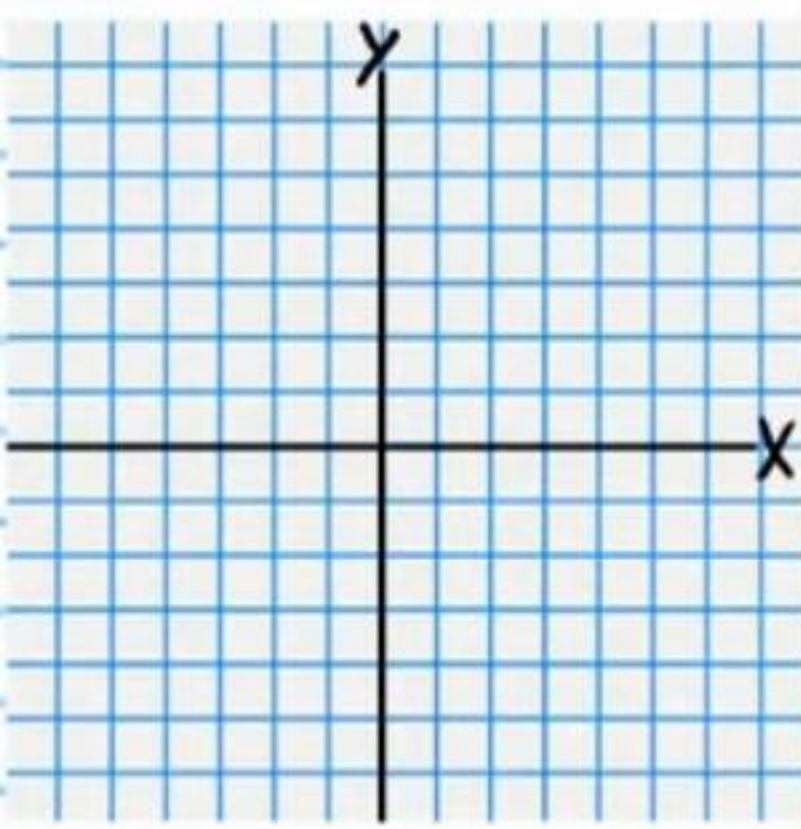
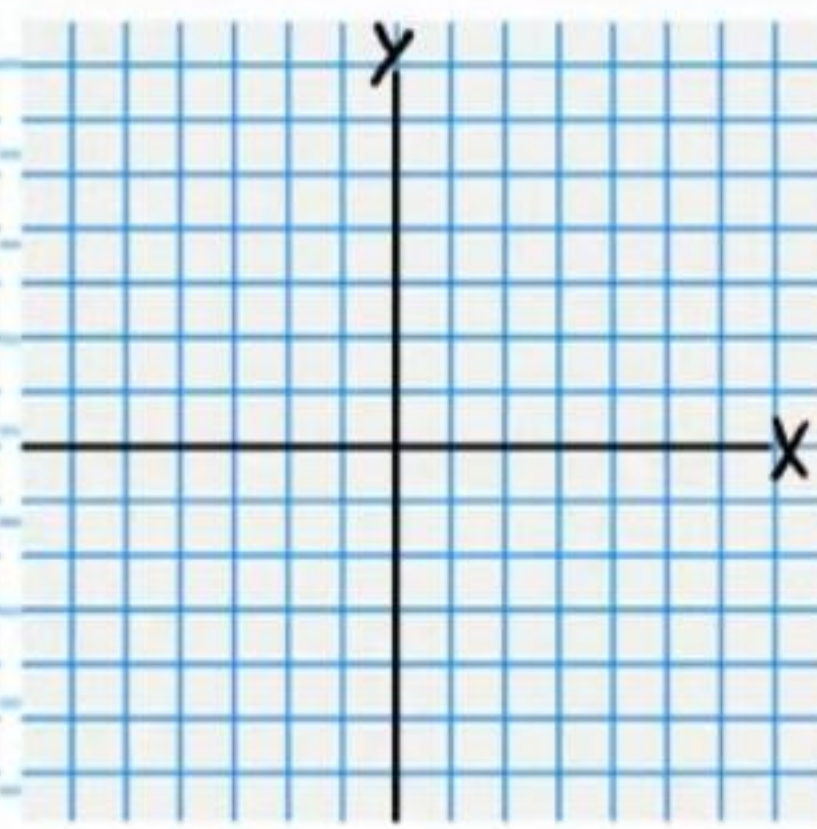
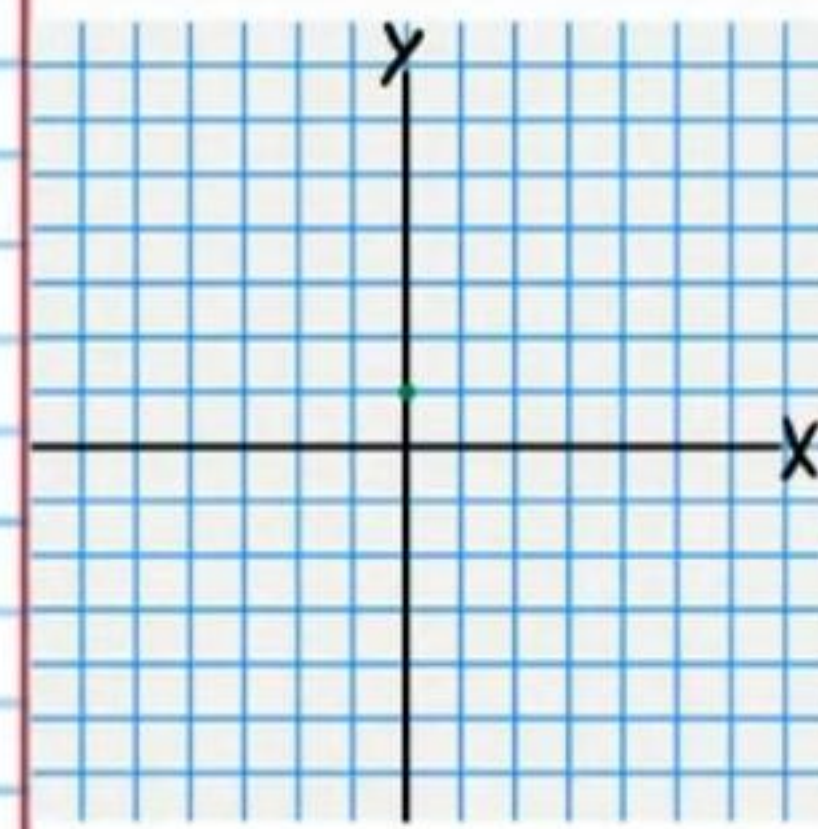


Altered Form:

Parent Relation

Transformation #1

Transformation #2



⑦ In the tables below, x represents the inputs and y represents the outputs. Circle the tables that could not represent functions.

x	6	8	10	0	4	-1
y	4	4	8	10	-2	1

x	-4	-2	0	2	-4	6
y	5	3	0	2	5	9

x	y
-6	7
-5	7
-4	7
-3	7
-2	7
-1	7

x	y
0	2
1	3
1	4
2	5
-1	6
1	7

⑧ If $f(x) = x^{\frac{3}{2}} + 2x$, find i) $f(1)$, ii) $f(4)$, iii) $f(b)$, and iv) $f(y+1)$.

i)

ii)

iii)

iv)

⑨ If $f(x) = 3^x - 1$ and $g(x) = \frac{x}{2}$, find the following:

i) $g \circ f(x)$

ii) $f \circ g(x)$

⑩ Given the function below, find the following.

$$f(x) = \begin{cases} \frac{1}{x}, & -5 \leq x \leq -1 \\ \sqrt{x}, & 0 < x \leq 7 \end{cases}$$

i) $f(-2)$

ii) $f(4)$

iii) $f(-\frac{1}{2})$

iv) $f(1)$

⑪ Write the domain of the function below in set notation.

$$h(x) = \log_4(2 - 3x)$$

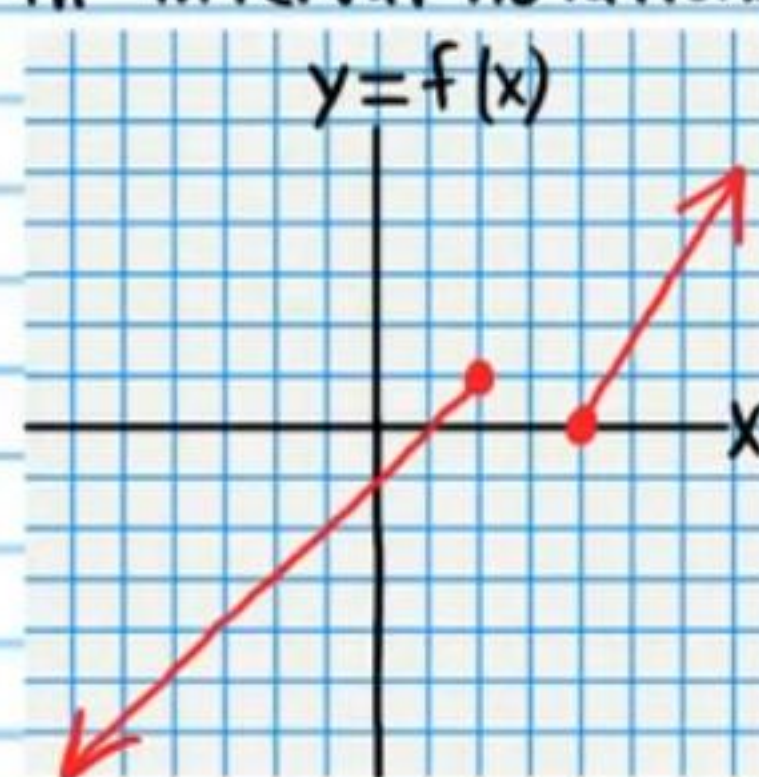
⑫ Write the domain of the function below in word form.

$$g(x) = -\frac{1}{5x}$$

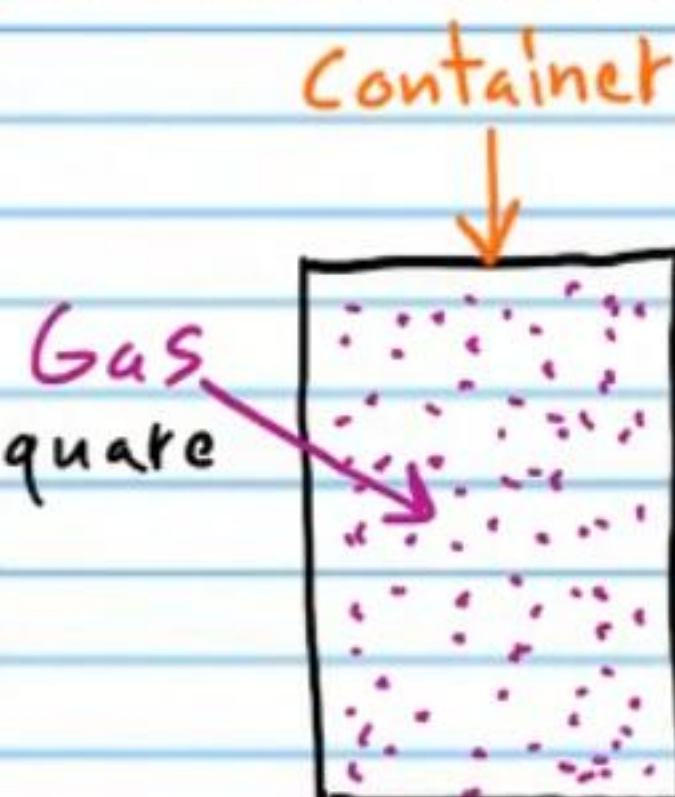
- ⑬ The graph of a function is written below. Write the domain and range of the function in interval notation.

Domain

Range



- ⑭ The pressure of the gas in the container at the right is directly proportional to the temperature of the gas, such that $P = 0.18T$, where P is pressure (in pounds per square inch) and T is temperature (in Kelvins).



a) If the pressure in the container is 41 psi, what is the temperature of the gas?

b) What is the proportionality constant?

- ⑮ If a 0.8 % interest rate is compounded continuously for 7 years, and the initial investment is \$9,921, how much would the investment be worth?

- ⑯ Tina received a student loan. If she does not make any payments on the loan for 15 years, and the debt grows from \$2,000 to \$14,000, what annual interest rate did the creditor charge if the interest is compounded quarterly?

Solve. Round to the nearest hundredth.

(17) $2 = \ln x^2 + 3 \ln x$ (18) $\frac{10^x}{10^{2x}} = 4$

(19) The number below represents the pH of a liquid.
Find the hydrogen ion concentration of the liquid.

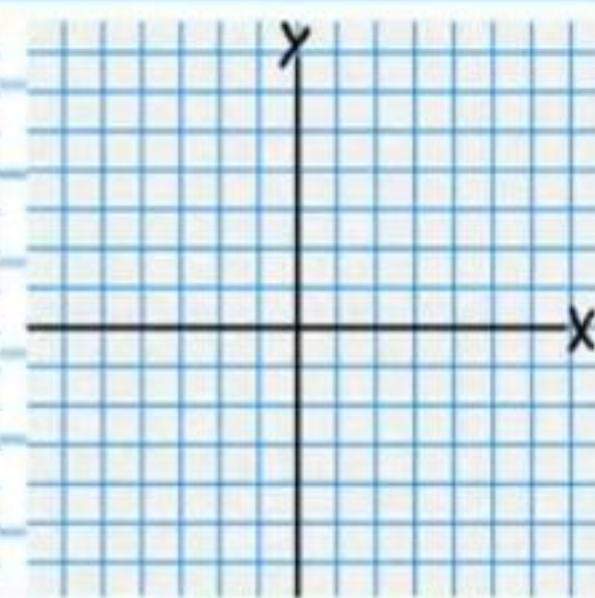
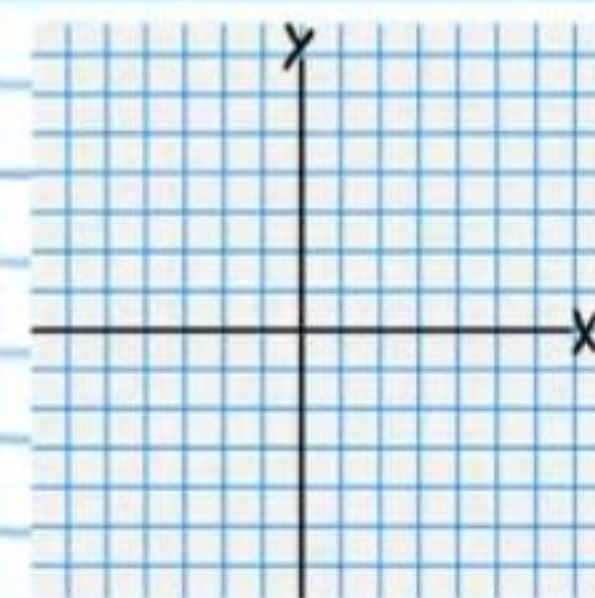
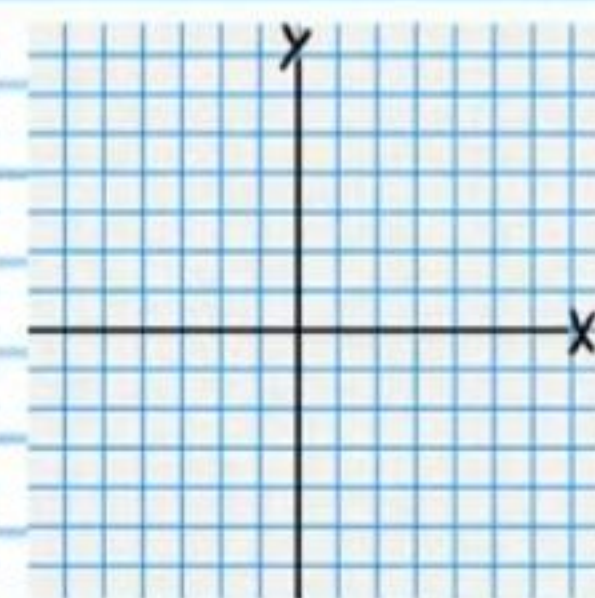
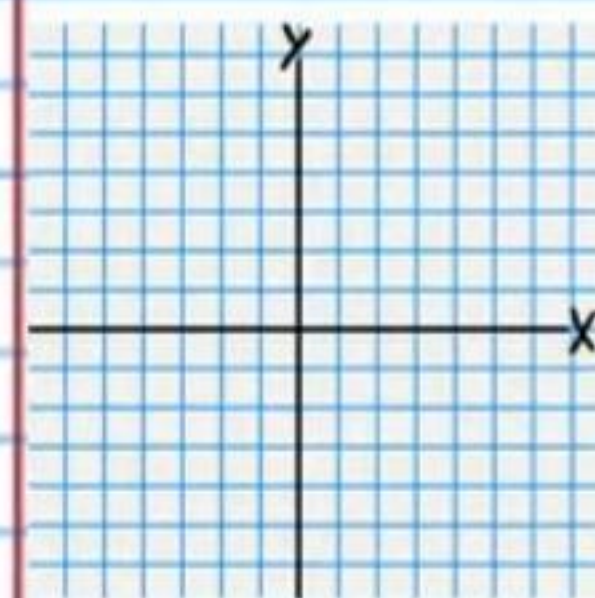
$\text{pH} = 6.7$

(20) Write the domain of the function below in set notation. You may use the grids below if necessary.

(21) Write the range of the function below in interval notation. You may use the grids below if necessary.

$f(x) = -4\sqrt{x-1} - 3$

Parent Function Transformation #1 Transformation #2 Final Transformation

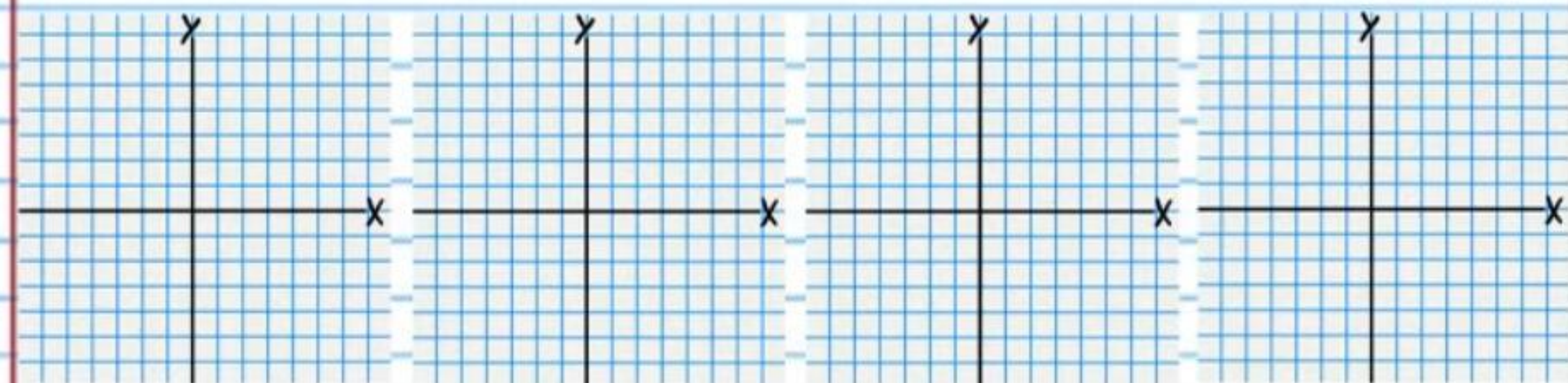


The transformation properties can be used with each pathway at the right to graph the corresponding equation. But in most of these pathways, expressions must be converted into a different form in order to use the transformation properties in the way that they are written. Choose the pathway(s) in which no modification of expressions is required. Then, use your chosen pathway to graph the corresponding equation.

$$\textcircled{2d} -y-2 = \log_3\left(\frac{x}{2}+2\right)$$

Pathway #1 $y = \log_3 x$ $y-2 = \log_3(x+2)$ $y-2 = \log_3\left(\frac{x}{2}+2\right)$ $-y-2 = \log_3\left(\frac{x}{2}+2\right)$	Pathway #2 $y = \log_3 x$ $y-2 = \log_3(x+2)$ $-y-2 = \log_3(x+2)$ $-y-2 = \log_3\left(\frac{x}{2}+2\right)$	Pathway #3 $y = \log_3 x$ $y = \log_3\left(\frac{x}{2}\right)$ $-y = \log_3\left(\frac{x}{2}\right)$ $-y-2 = \log_3\left(\frac{x}{2}+2\right)$
Pathway #4 $y = \log_3 x$ $y = \log_3\left(\frac{x}{2}\right)$ $y-2 = \log_3\left(\frac{x}{2}+2\right)$ $-y-2 = \log_3\left(\frac{x}{2}+2\right)$	Pathway #5 $y = \log_3 x$ $-y = \log_3 x$ $-y = \log_3\left(\frac{x}{2}\right)$ $-y-2 = \log_3\left(\frac{x}{2}+2\right)$	Pathway #6 $y = \log_3 x$ $-y = \log_3 x$ $-y-2 = \log_3(x+2)$ $-y-2 = \log_3\left(\frac{x}{2}+2\right)$

Parent Relation Transformation #1 Transformation #2 Final Transformation

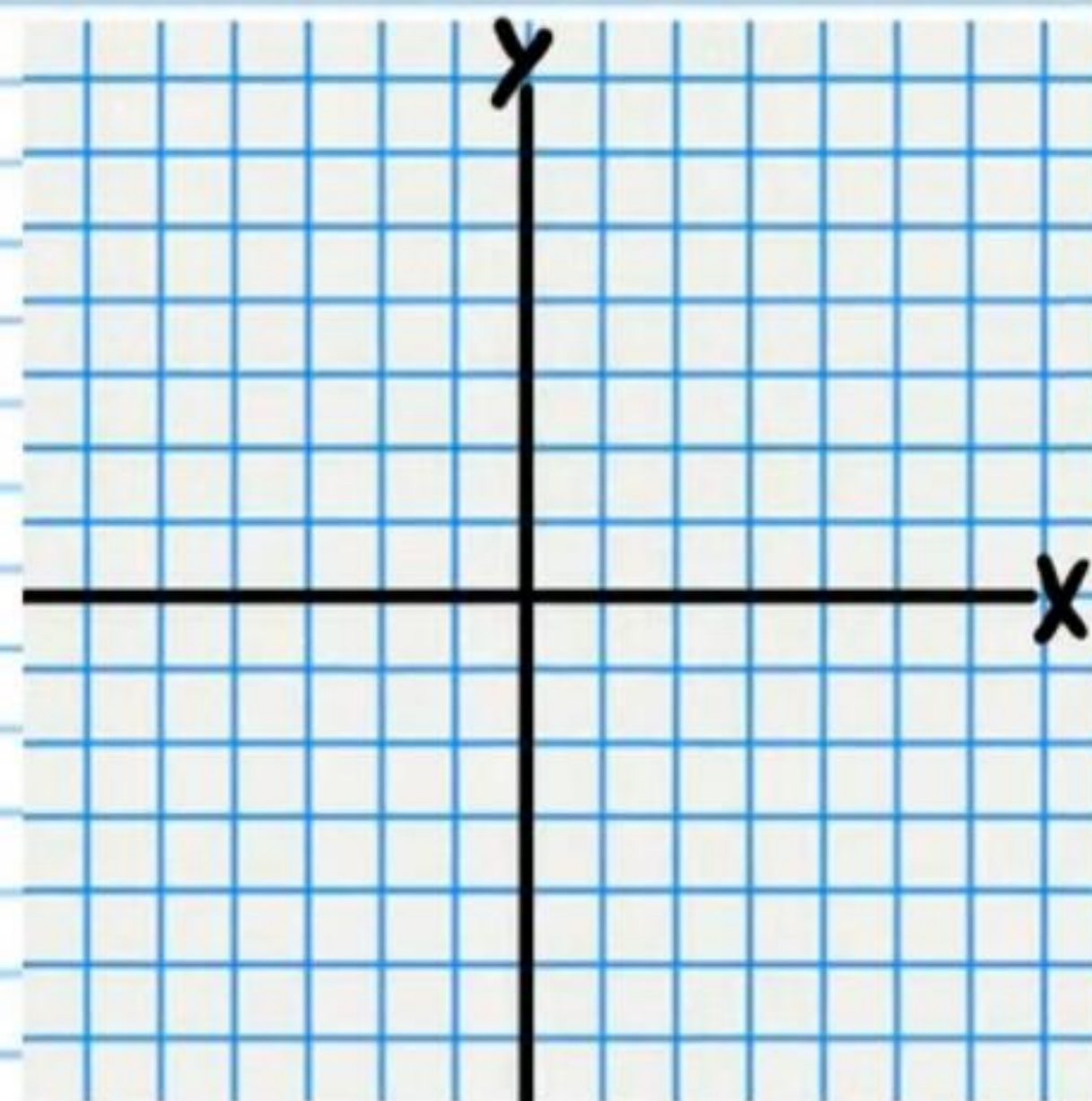


②③ Write the parent relation and the parent equation of the equation below. Then, graph the PARENT relation.

Parent Relation:

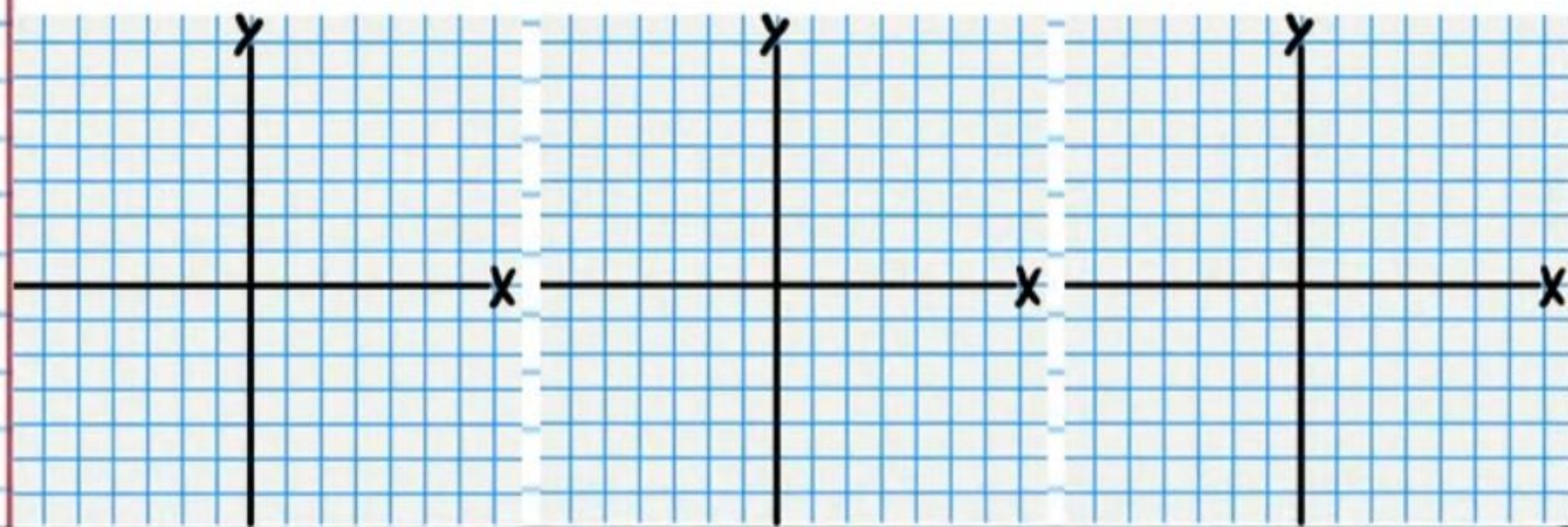
Parent Equation:

$$y = \frac{4^{-(x+2)}}{3} - 1$$



②④ Graph the equation above by performing multiple transformations on the parent graph. Draw the asymptote(s) as dotted lines.

Transformation #1 Transformation #2 Final Graph

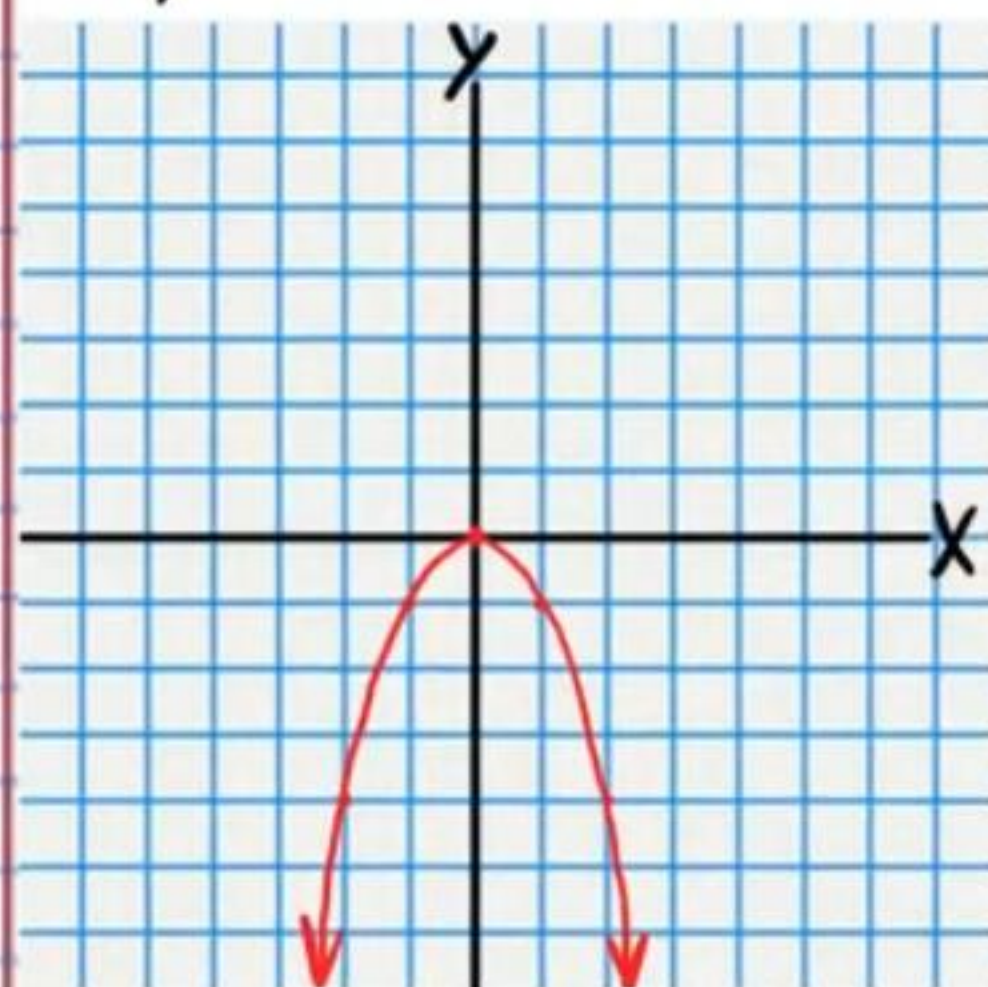


②⑤ There are 39 coins in a fountain, no pennies, only quarters, dimes, and nickels. If the number of nickels is twice the number of dimes, and the total value of the coins is \$6.45, how many of each type of coin are there?

Answers

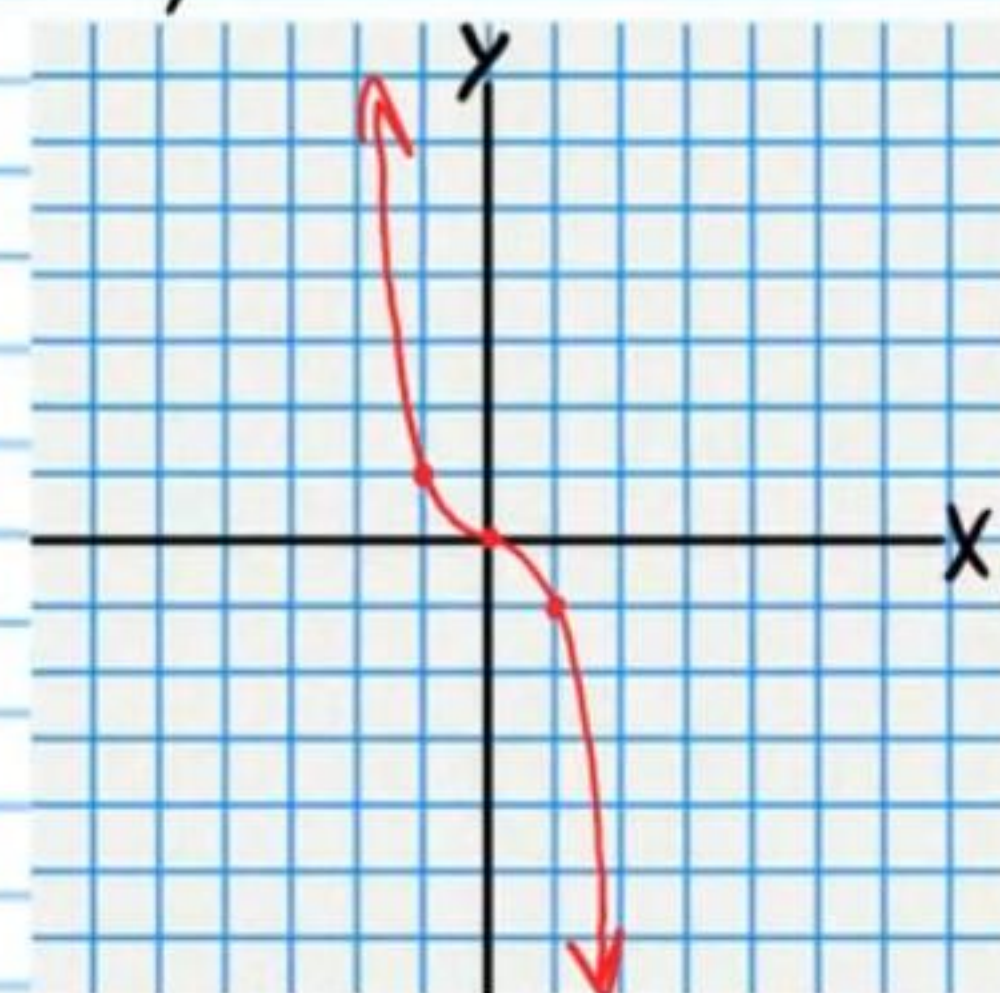
① Quadratic

$$y = x^2$$



② Cubic

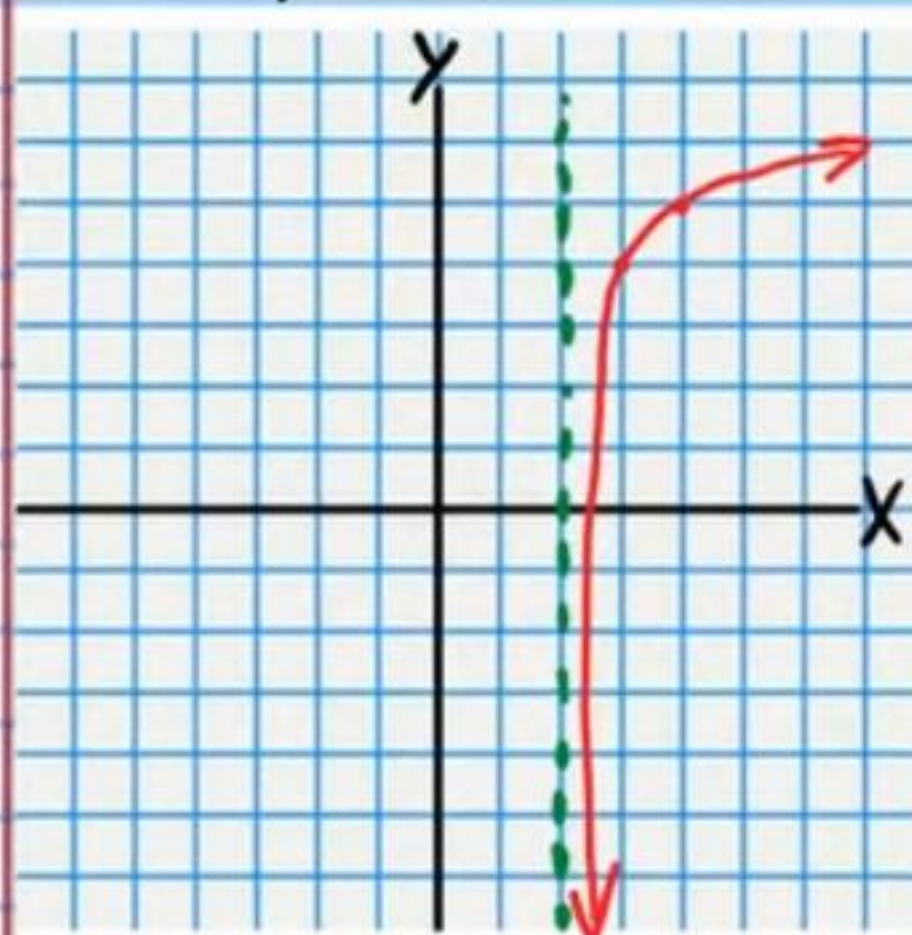
$$x^3 = y$$



③ Logarithmic

$$y = \log_2 x$$

$$x = 2$$

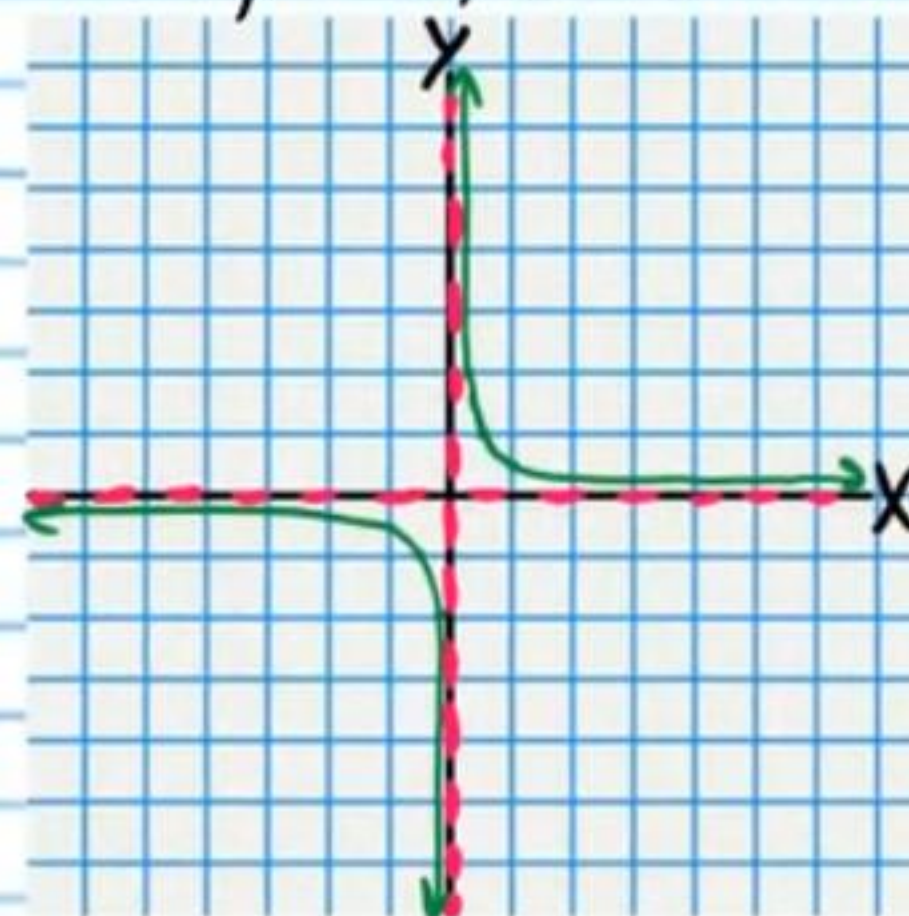


④ Reciprocal

$$y = \frac{1}{x}$$

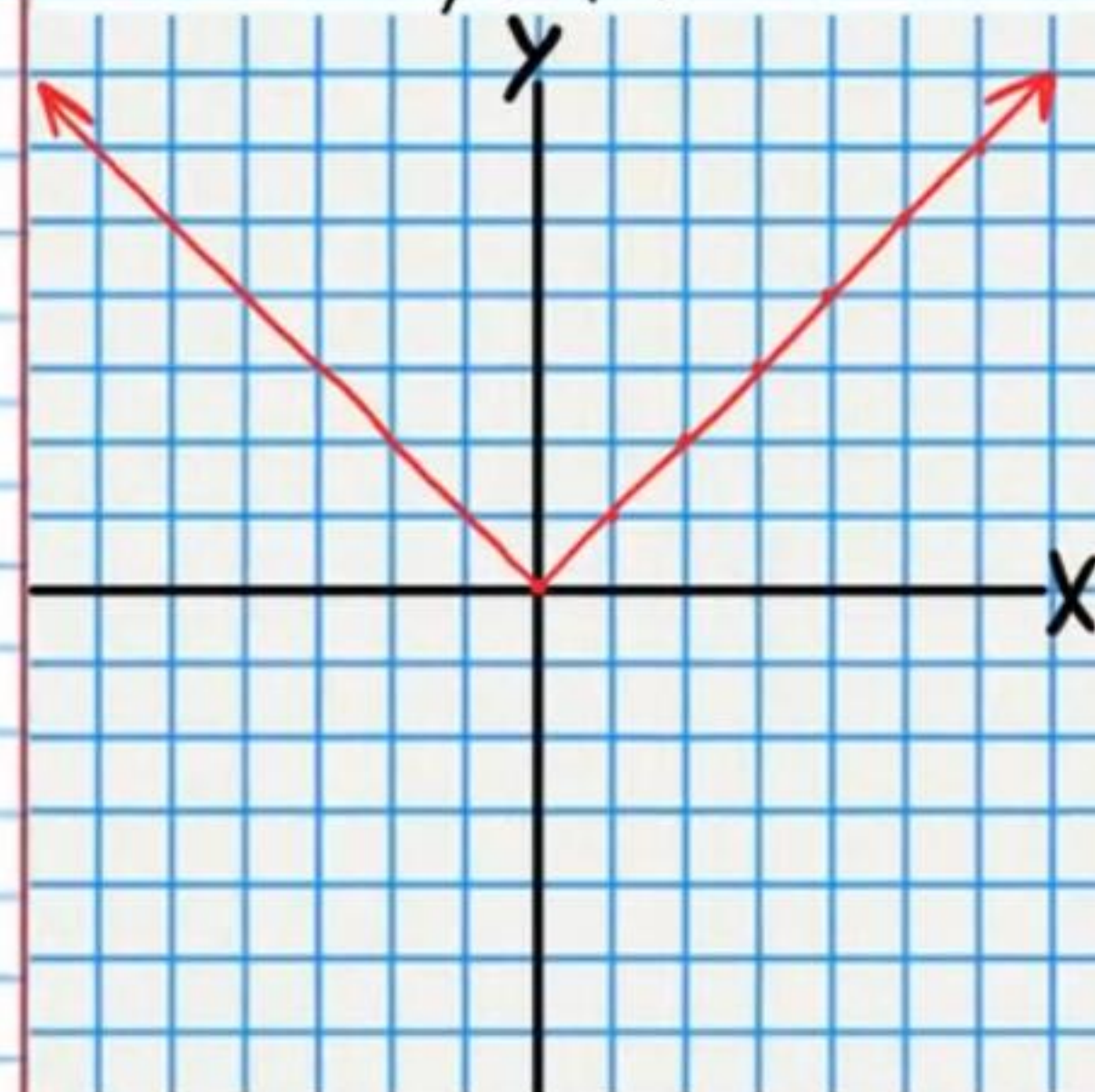
Horizontal compression

$$y = 0, x = 0$$

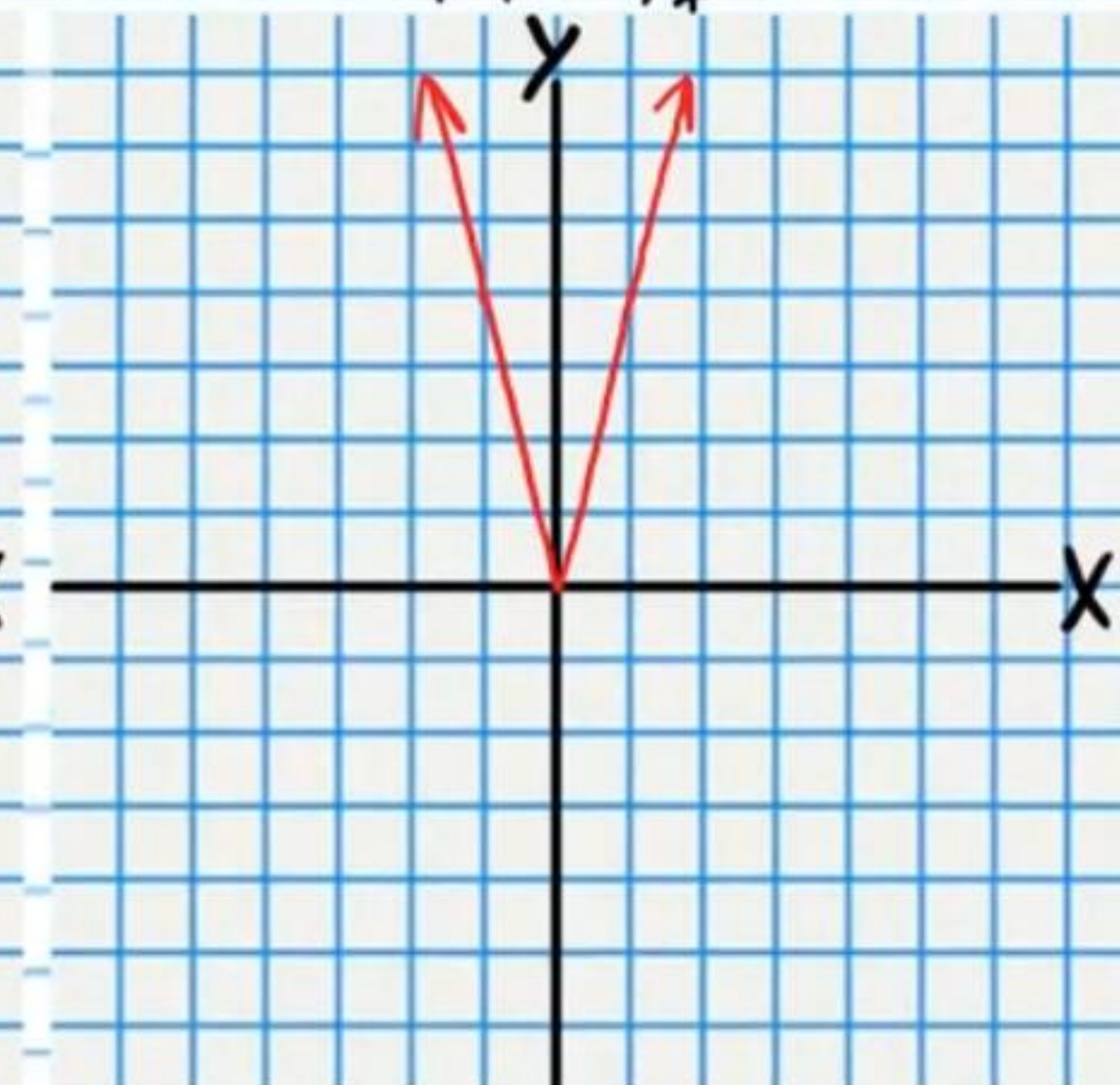


⑤ $|x| = \frac{y}{4}$

Parent Relation
 $y = |x|$

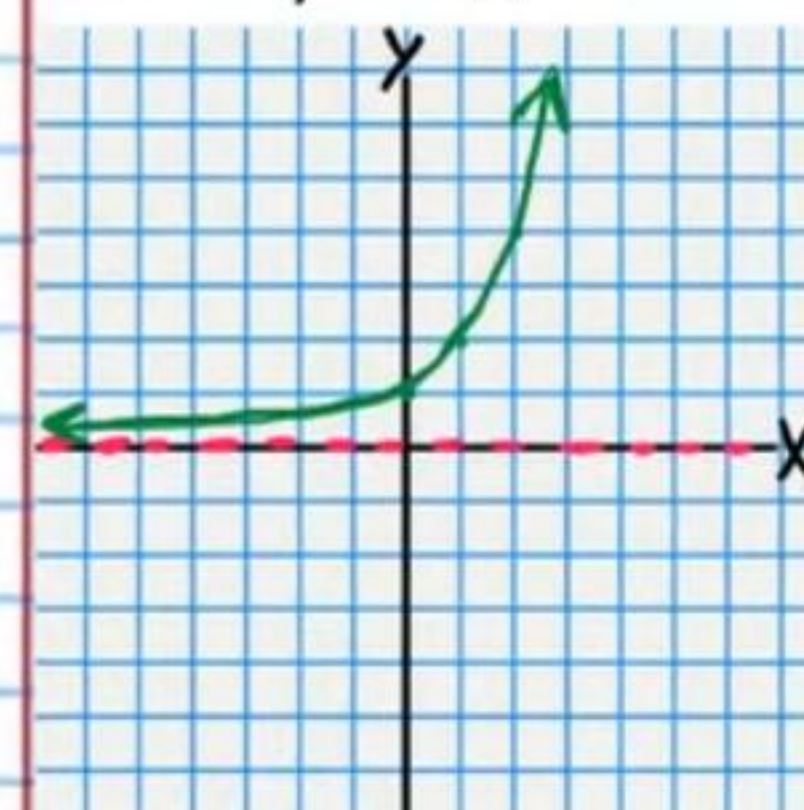


Transformation
 $|x| = \frac{y}{4}$

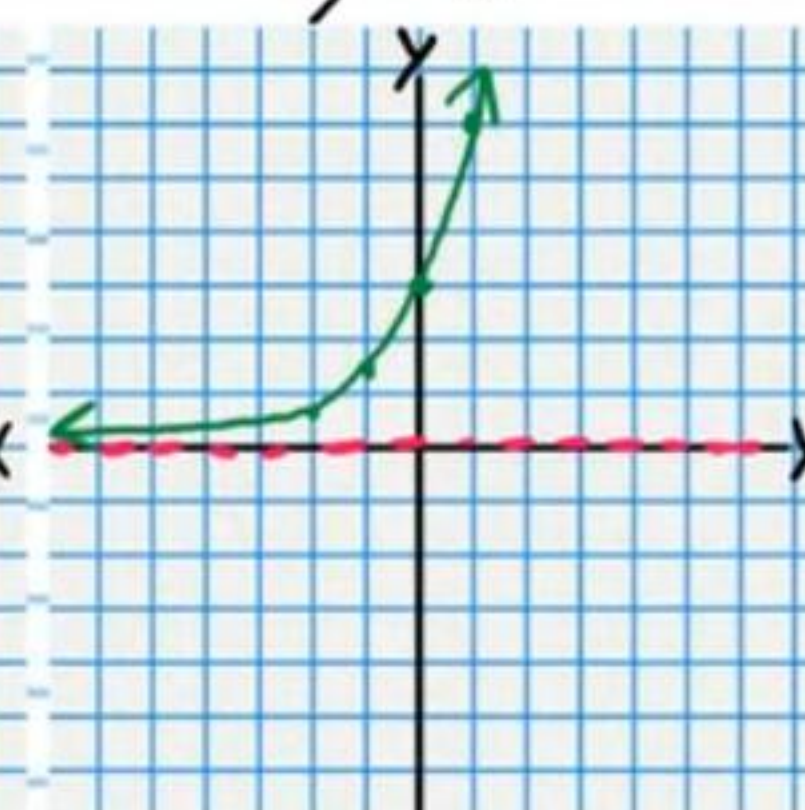


⑥ $\frac{y+3}{3} = 2^{x+3}$

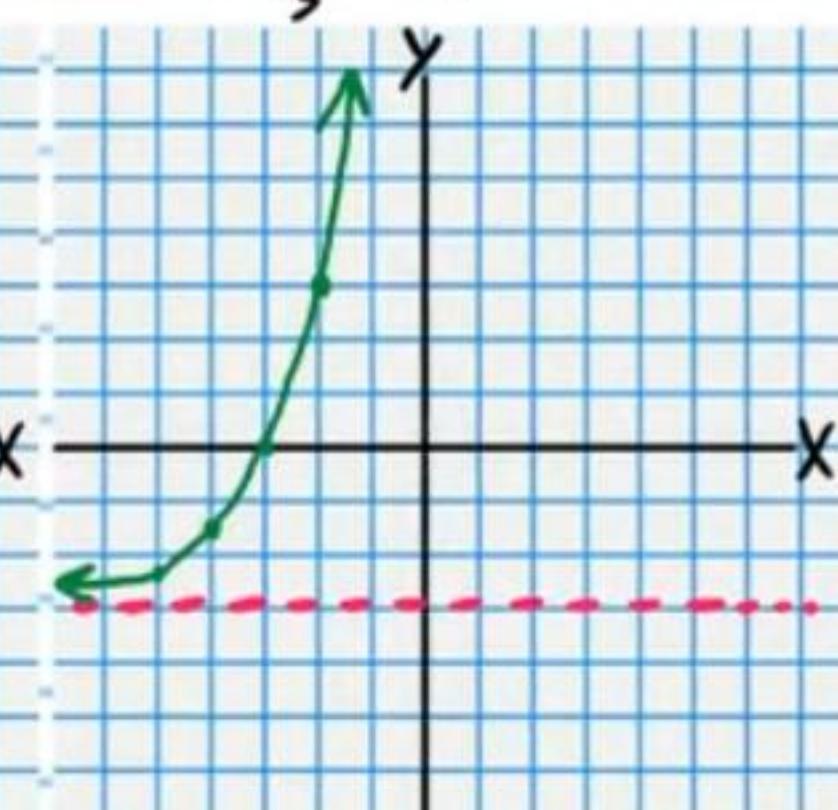
Parent Relation
 $y = 2^x$



Transformation #1
 $\frac{y}{3} = 2^x$



Transformation #2
 $\frac{y+3}{3} = 2^{x+3}$



⑦

X	6	8	10	0	4	-1
Y	4	4	8	10	-2	1

X	-4	-2	0	2	-4	6
Y	5	3	0	2	5	9

X	Y
-6	7
-5	7
-4	7
-3	7
-2	7
-1	7

X	Y
0	2
1	3
1	4
2	5
-1	6
1	7

⑧ i) $\boxed{3}$

ii) $\boxed{16}$

iii) $\boxed{\sqrt{b^3} + 2b}$ or $\boxed{(\sqrt{b})^3 + 2b}$

iv) $\boxed{\sqrt{(y+1)^3} + 2(y+1)}$ or $\boxed{(\sqrt{y+1})^3 + 2(y+1)}$

⑨ i) $\boxed{\frac{3^x - 1}{2}}$

ii) $\boxed{(3^x - 1)\left(\frac{x}{2}\right)}$

⑩ i) $\boxed{\frac{1}{-2}}$

iii) $\boxed{\text{Undefined}}$

ii) $\boxed{2}$

iv) $\boxed{1}$

⑪ $\boxed{\{x \mid x < \frac{2}{3}\}}$

⑫ $\boxed{\text{All real numbers except zero.}}$

⑬ Domain: $\boxed{(-\infty, 2] \cup [4, \infty)}$

Range: $\boxed{(-\infty, \infty)}$

⑭ a) $\boxed{T = 227.7 \text{ Kelvins}}$ b) $\boxed{0.18}$

⑮ $\boxed{\$10,492.43}$

⑯ $\boxed{13.185\%}$

⑰ $\boxed{x \approx 1.49}$

⑱ $\boxed{x \approx -0.6}$

⑲ $\boxed{0.0000001995 \text{ moles/liter} \approx [H^+]}$

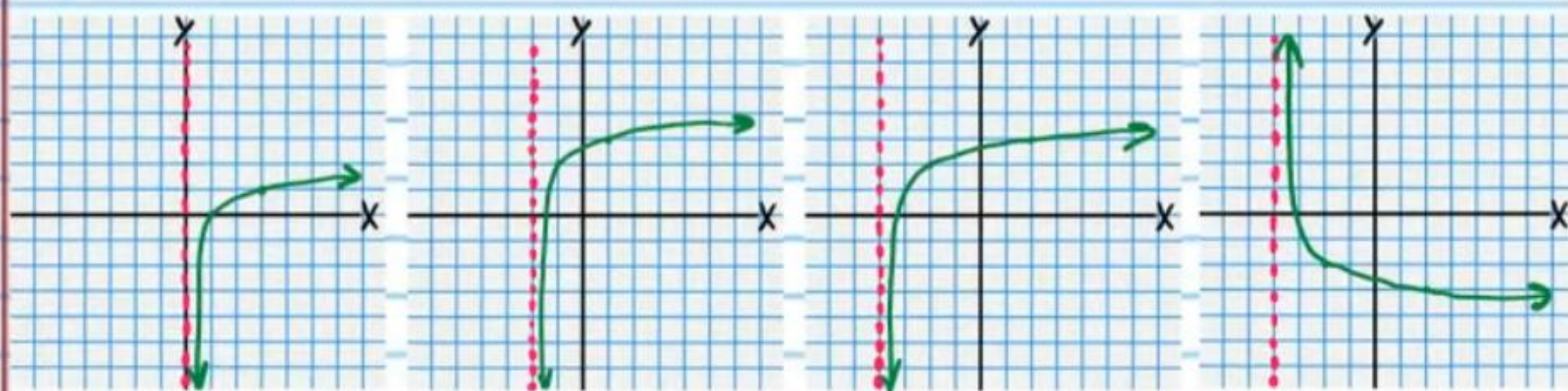
⑳ $\boxed{\{x \mid x \geq 1\}}$

㉑ $\boxed{(-\infty, -3]}$

② Pathway #1 or #2.

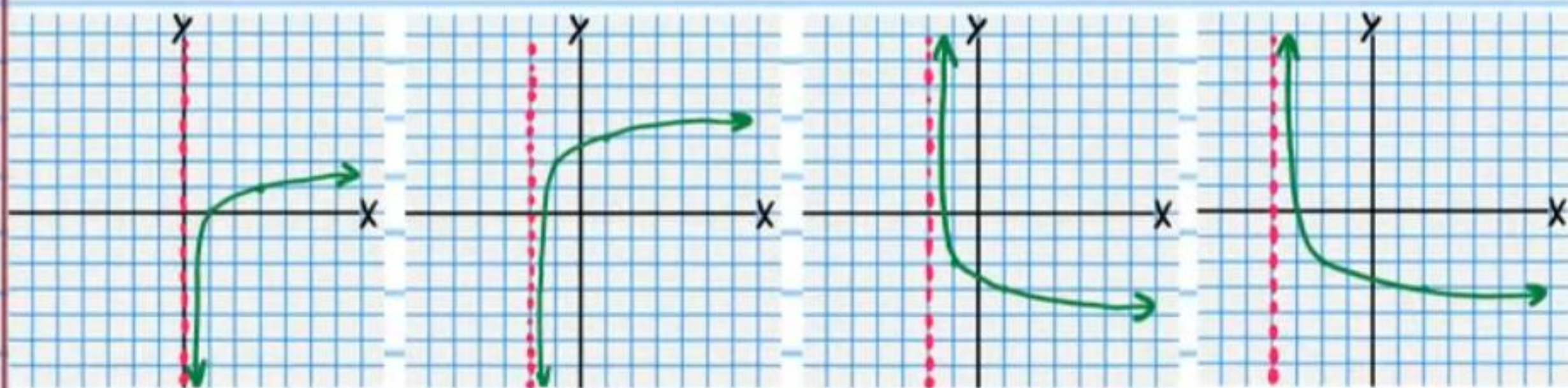
Pathway #1

Parent Relation	Transformation #1	Transformation #2	Final Transformation
$y = \log_3 x$	$y - 2 = \log_3(x + 2)$	$y - 2 = \log_3(\frac{x}{2} + 2)$	$-y - 2 = \log_3(\frac{x}{2} + 2)$



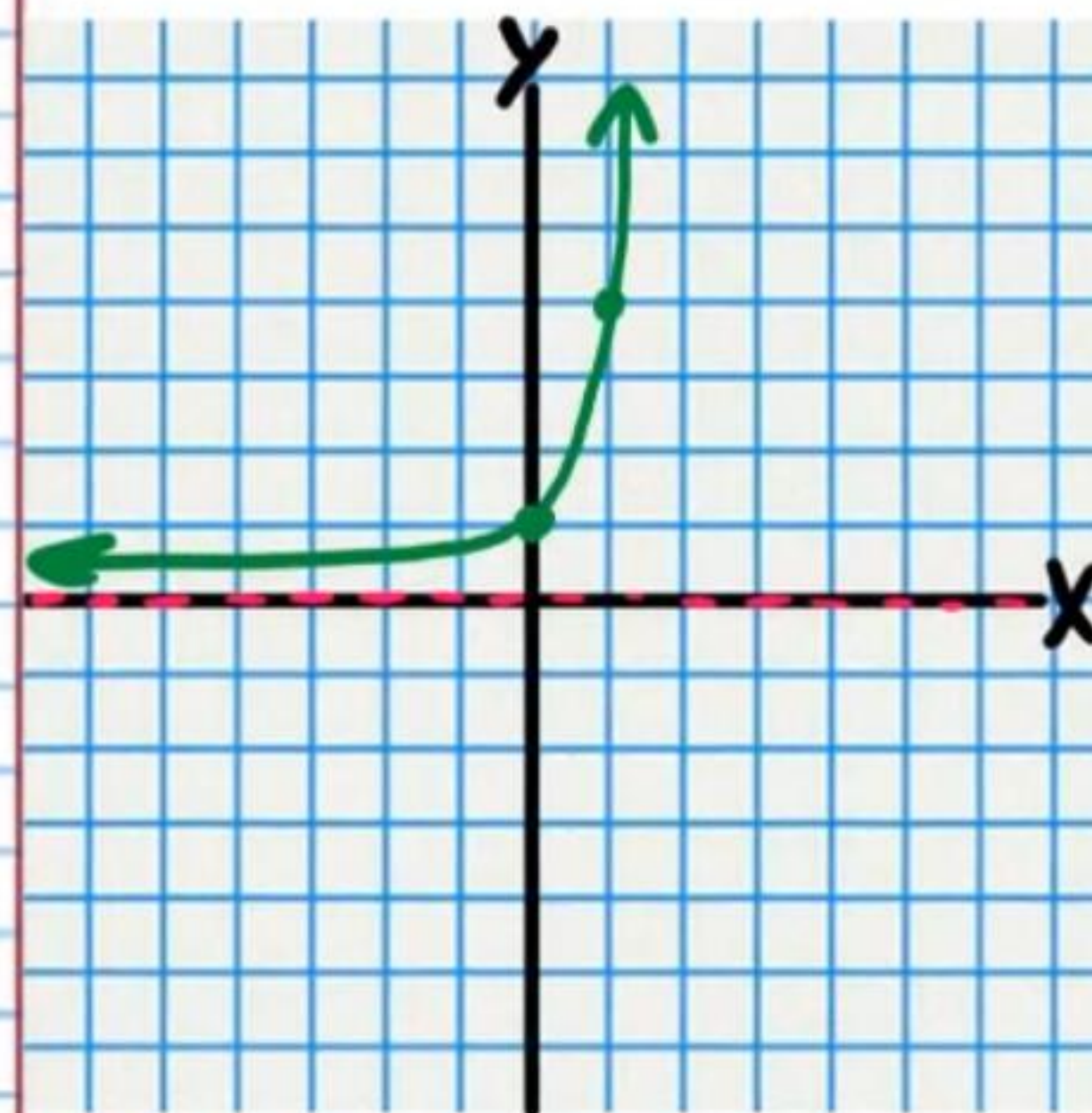
Pathway #2

Parent Relation	Transformation #1	Transformation #2	Final Transformation
$y = \log_3 x$	$y - 2 = \log_3(x + 2)$	$-y - 2 = \log_3(x + 2)$	$-y - 2 = \log_3(\frac{x}{2} + 2)$



②3 Exponential

$$y = 4^x$$

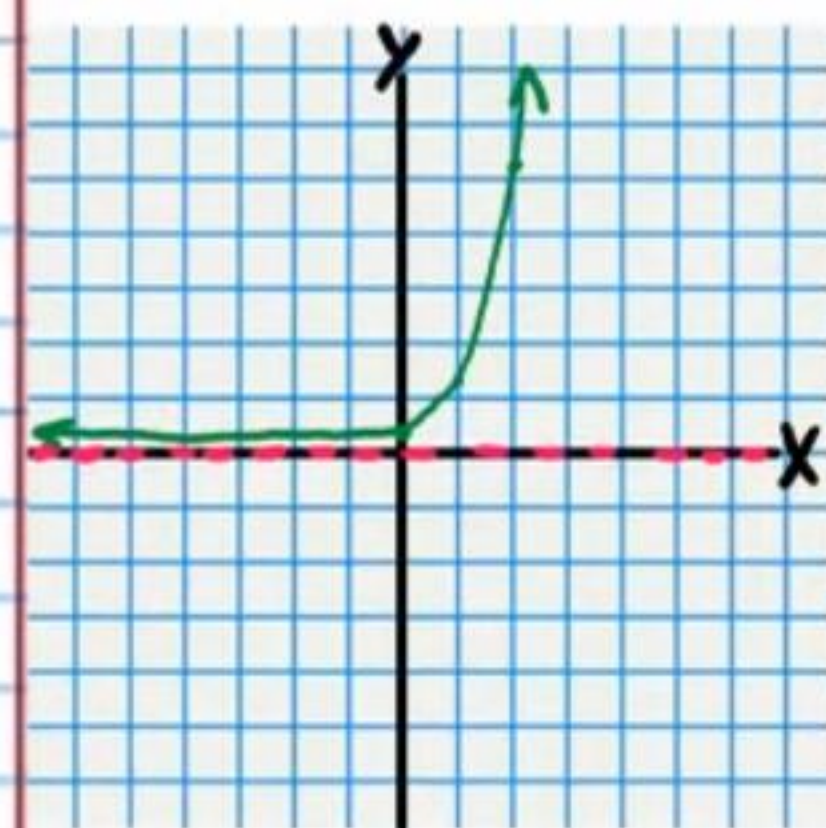


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Option #1

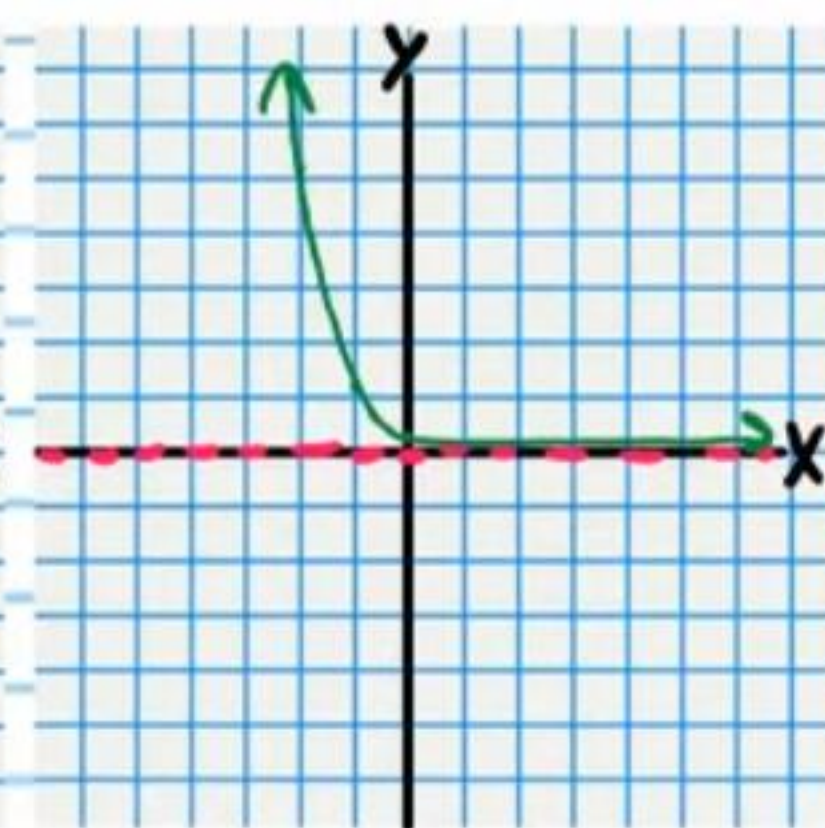
Transformation #1

$$3y = 4^x$$



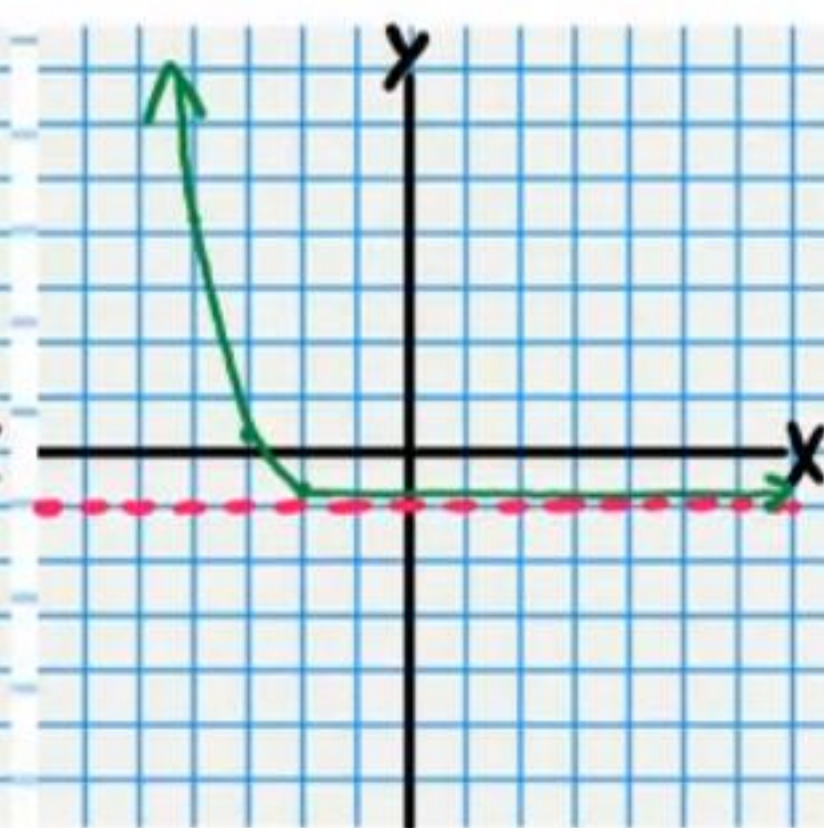
Transformation #2

$$3y = 4^{-x}$$



Final Graph

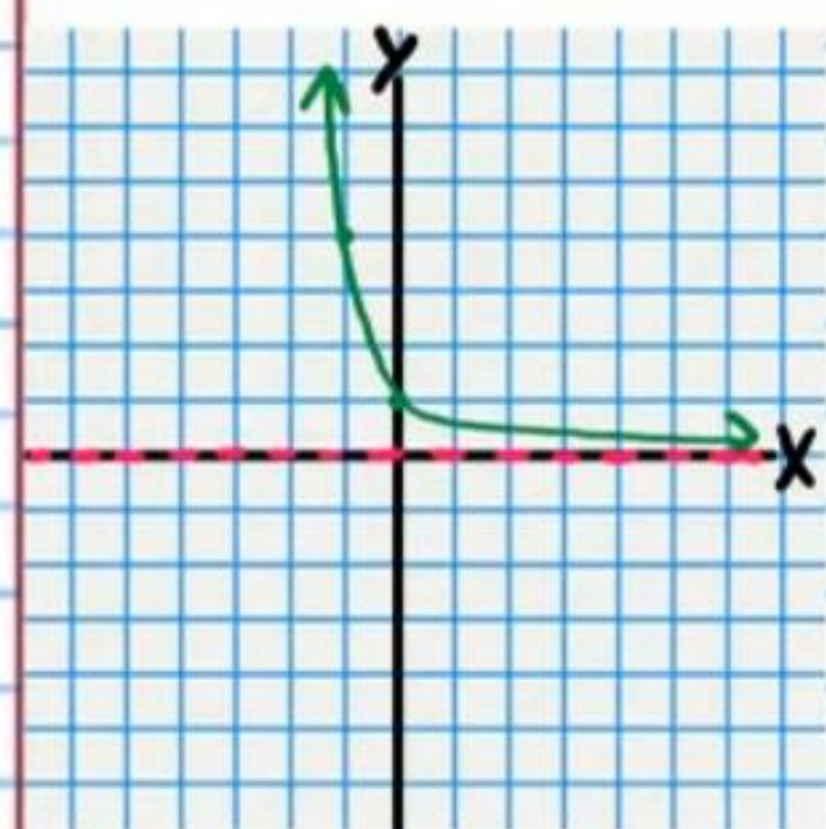
$$3(y+1) = 4^{-(x+2)}$$



Option #2

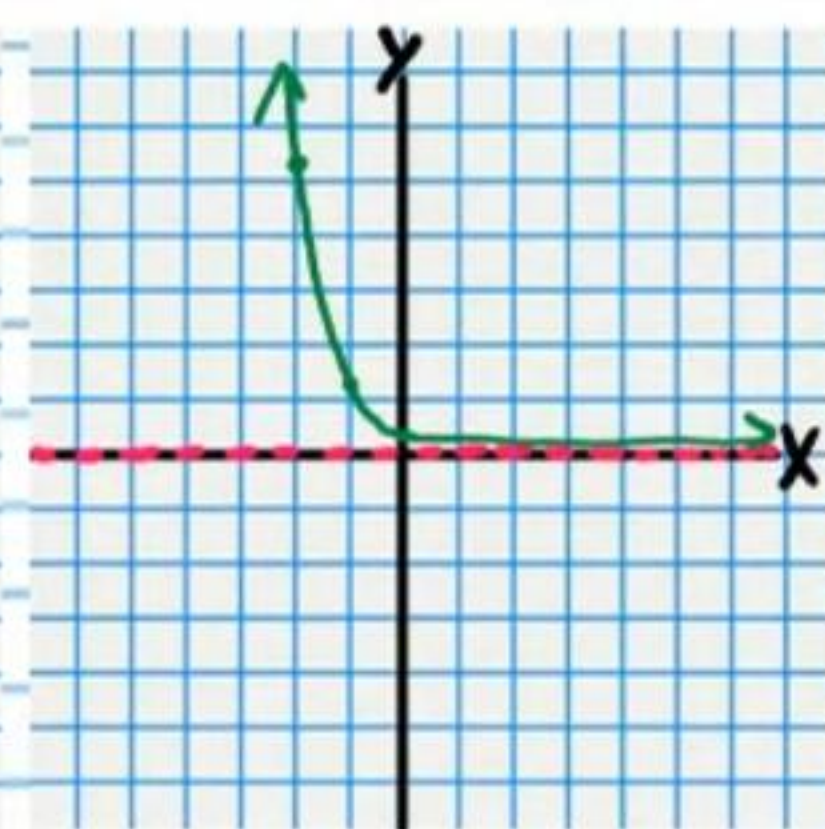
Transformation #1

$$y = 4^{-x}$$



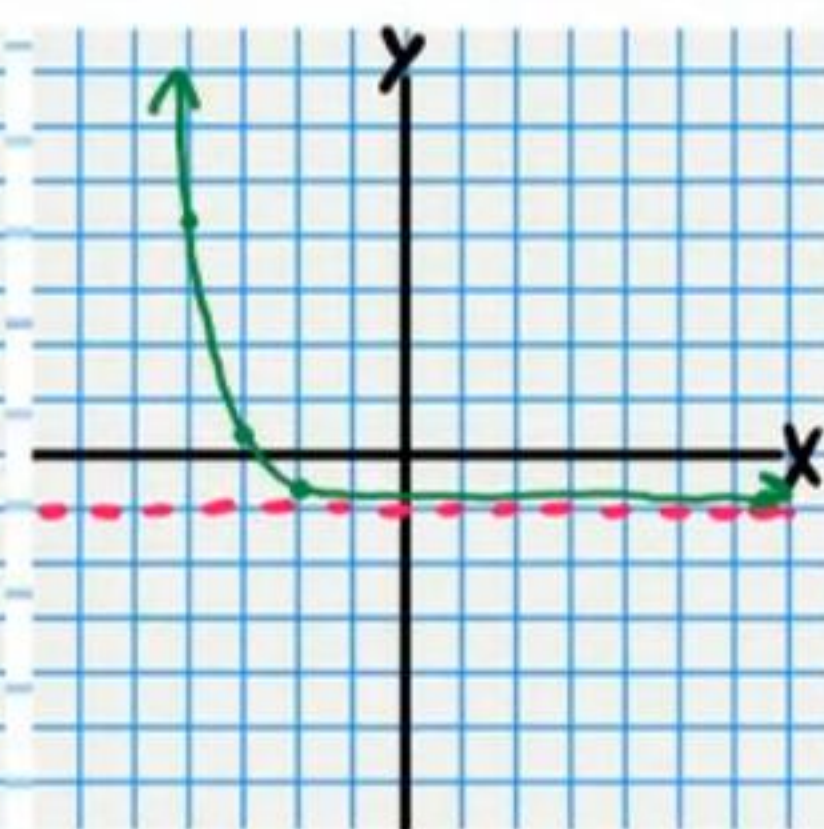
Transformation #2

$$3y = 4^{-x}$$



Final Graph

$$3(y+1) = 4^{-(x+2)}$$



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21 quarters, 12 nickels, 6 dimes